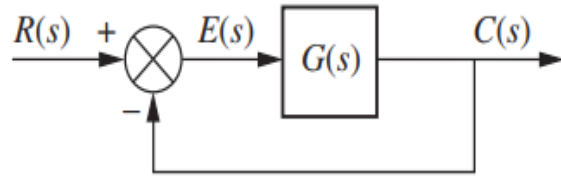


Control Systems (EEE/ECE/INSTR F242)

Tutorial 9 (17 March 2026)

- Given the unity feedback system of Fig. Q1 with $G(s) = \frac{K(s+5)}{s(s+1.2)(s+3)}$, find the following
 - The range of K that keeps the system stable.
 - The value of K that makes the system oscillate.
 - The frequency of oscillation when K is set to the value that makes the system oscillate.



- For the polynomial $S^6 + 3s^5 + 4s^4 + 6s^3 + 5s^2 + 3s + 2$, determine the number of poles in the **left-half plane (LHP)**, **right-half plane (RHP)**, and on the **imaginary axis (IAP)**. Additionally, determine the **values of all poles and indicate their locations in the s-plane**.
- Find the value of K in the system of Fig. Q2(a) that will place the closed-loop poles as shown in Fig. Q2(b).

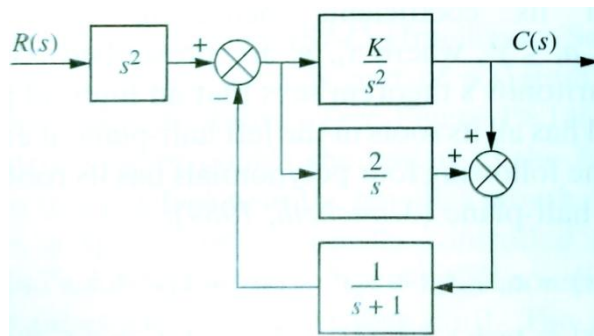


Fig. Q2(a)

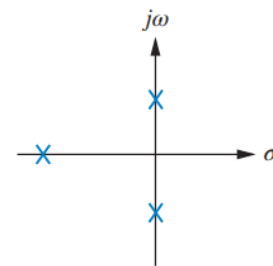


Fig. Q2(b)

- For the closed-loop system shown below in Fig. Q3:
 - Determine the value of α and K for the overall system to be stable.
 - Suggest suitable values of α and K , while assuring that the steady-state error due to a unit ramp input is no more than 20%.

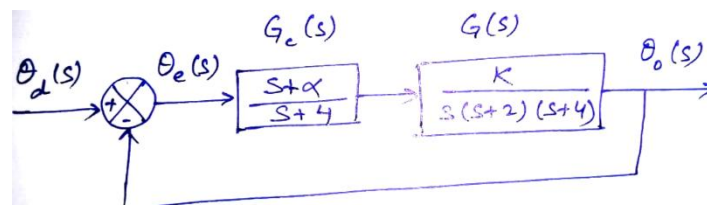


Fig. Q3